# CaseFlow AI

## Product Requirements Document (PRD)

**Version:** 2.0 — Survey-Informed MVP
**Date:** October 22, 2026
**Status:** Ready for MVP Prototyping
**Product Category:** AI-Powered Academic Collaboration / EdTech
**Initial Target Market:** Business Analytics Students and Case Competition Teams

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# 1. Executive Summary

**CaseFlow AI** is an AI-powered project kickoff platform designed to help analytics students organize group projects more effectively.

The application allows teams to upload an assignment or case competition prompt, receive three suggested work allocations, and coordinate their first meeting through a shared availability scheduler.

Unlike traditional project management tools, CaseFlow AI emphasizes **analytics-aware task allocation, skill matching, workload fairness, and meeting coordination**.

The product is informed by survey research involving 41 USC Marshall MSBA students. The findings indicate that workload balance, matching tasks to individual strengths, and conflicting schedules are relevant coordination challenges.

### Core Value Proposition

**"From assignment prompt to coordinated team in minutes."**

CaseFlow AI helps students make better decisions about who should do what and when the team should meet, without requiring extensive manual planning.

### MVP Features

1. **Analytics-Aware Work Allocation:** Interpret project requirements and suggest three editable work distributions.

2. **Collaborative Meeting Scheduler:** Collect team availability and identify overlapping meeting times.

The MVP will prioritize these two features rather than expanding into full project management.

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# 2. Market Research & Problem Validation

## 2.1 Primary Research

**Research Source:** USC Marshall MSBA B7 Case Competition Survey
**Sample Size:** 41 respondents
**Research Method:** Student questionnaire
**Target Audience:** Students with experience working on collaborative academic projects and competitions

The research explored how teams divide responsibilities, what challenges they encounter, how they schedule meetings, and their interest in a proposed planning and scheduling application.

## 2.2 Key Findings

| Survey Finding | Result | Product Implication |
|---|---|---|
| Students experience difficulty balancing workloads | 53.7% (22/41) | Prioritize workload visibility |
| Students prefer strength-based task allocation | 43.9% (18/41) | Make skill-based allocation the default |
| Students prefer balanced workload allocation | 29.3% (12/41) | Provide balanced allocation as an alternative |
| Students report conflicting meeting schedules | 61.0% (25/41) | Include an availability overlap scheduler |
| Students currently use group chats to coordinate meetings | 87.8% (36/41) | Reduce scheduling back-and-forth |
| Students would be somewhat or very likely to try the app | 39.0% (16/41) | Validate actual adoption through pilot testing |

### Additional Findings

- 56.1% (23/41) described agreeing on work allocation as somewhat or very easy.
- 41.5% (17/41) identified understanding assignment requirements as a challenge.
- 39.0% (16/41) identified breaking projects into clear tasks as a challenge.
- 68.3% (28/41) agreed on their most recent meeting time within one hour.
- 31.7% (13/41) were neutral about trying the application, while 29.3% (12/41) were unlikely to try it.

## Research Interpretation

**Finding A: The problem is not simply dividing responsibilities.**

Most respondents do not describe the initial division of work as particularly difficult.

The stronger opportunity is improving the *quality* of task allocation by accounting for workload fairness and relevant skills.

**Finding B: Strength-based recommendations offer differentiation.**

Strength-based allocation was the most popular of the three proposed strategies.

This supports creating an application that recognizes analytical skills such as Python, SQL, statistical analysis, visualization, and business storytelling.

**Finding C: Meeting scheduling is a widespread coordination challenge.**

Conflicting schedules were the most common reported scheduling issue.

However, many teams already agree on meeting times relatively quickly. The product must demonstrate greater convenience than existing group chats and scheduling polls rather than assuming substantial delays affect every team.

**Finding D: Adoption is not yet validated.**

Only 39% of respondents expressed positive intent to try the application. The MVP must establish whether students find the integrated workflow valuable enough to change their existing habits.

## Research Limitations

The sample is small and drawn from a specific academic population. Results should be treated as exploratory and not generalized to all university students.

The work-allocation challenge question also contains three submissions exceeding its stated three-choice limit. These findings inform prioritization but do not constitute conclusive market validation.

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# 3. Problem Statement

Students participating in business analytics projects and case competitions frequently need to translate complex assignment instructions into a coordinated team plan.

Although teams can often agree on who handles each responsibility, they may struggle to ensure that:

- Workloads are reasonably balanced.
- Tasks match individual technical strengths.
- Required analytical deliverables are clearly identified.
- Dependencies between analytical workstreams are understood.
- Meeting times accommodate competing student schedules.

Currently, students address these problems through combinations of assignment documents, group chats, ChatGPT, and scheduling tools.

**The opportunity is not to replace these tools individually, but to connect project interpretation, intelligent work allocation, and meeting coordination into a single kickoff workflow.**

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# 4. Target Users

## Primary Audience

Undergraduate and graduate students participating in:

- Business analytics group assignments
- Data science projects
- Case competitions
- Statistics and machine learning assignments
- Consulting-oriented analytical projects

## Secondary Audience

Future potential users include hackathon participants, student consulting organizations, academic research teams, and other small project-based teams.

## Primary User Persona

**Persona: Graduate Analytics Student**

A five-person analytics team receives a case competition prompt requiring data analysis, business recommendations, and a final presentation.

Team members have different strengths in Python, SQL, statistics, financial analysis, and business communication.

The team needs to interpret the requirements, distribute responsibilities, and identify a mutually available kickoff meeting time.

### Jobs to Be Done

1. Understand what the assignment actually requires.
2. Divide analytical responsibilities fairly.
3. Match technical and nontechnical work to relevant skills.
4. Establish an initial work plan everyone understands.
5. Identify a meeting time without excessive messaging.

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# 5. Product Goals

### Goal 1: Improve Work Allocation Quality

Generate allocations that account for analytical requirements, workload balance, and team capabilities.

### Goal 2: Accelerate Initial Planning

Reduce the time needed to produce an acceptable initial project plan.

### Goal 3: Simplify Meeting Coordination

Provide an intuitive method for collecting availability and identifying feasible meeting times.

### Goal 4: Minimize User Friction

Allow teams to begin planning with minimal setup and avoid requiring accounts for scheduling participants.

### Goal 5: Validate Product Demand

Determine whether analytics students find sufficient value in the combined workflow to use it for future assignments.

## Non-Goals for MVP

The initial application will not include:

- Full project management or task tracking
- Automated analytical modeling
- Dataset processing or visualization
- Automatic project completion monitoring
- Google Calendar or Outlook integrations
- Team messaging
- Automated reminders
- Assignment grading or submission evaluation
- Advanced project performance dashboards

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# 6. Core Feature 1: Analytics-Aware Work Allocation

**Priority: P0 — Core MVP**

## 6.1 Feature Description

Users upload or paste their assignment instructions and enter team member information.

The AI interprets the prompt, identifies required deliverables, and generates three alternative methods for allocating responsibilities.

The application is specifically designed to recognize analytical project requirements rather than produce generic task lists.

## 6.2 Required Inputs

- Assignment instructions (pasted text or supported document upload)
- Team member names

# Optional Inputs

- Project deadline
- Member skills
- Member preferences
- Member experience levels

### Supported Skill Categories

The MVP may recognize skills including:

- Python
- SQL
- Statistics
- Machine learning
- Data cleaning and preparation
- Data visualization
- Business strategy
- Financial analysis
- Research
- Presentation and storytelling

Users should also be able to enter their own skills.

# 6.3 Assignment Interpretation

The system extracts major deliverables and identifies the analytical workstreams needed to complete them.

### Potential Analytical Workstreams

| Workstream | Example Responsibility |
|---|---|
| Problem Framing | Define business objectives and project KPIs |
| Data Preparation | Clean data and validate dataset quality |
| Exploratory Data Analysis | Analyze patterns and trends |
| Statistical Analysis | Perform hypothesis testing or regression |
| Predictive Modeling | Develop and evaluate models |
| Business Strategy | Interpret findings and develop recommendations |

| | |
|---|---|
| Presentation | Prepare visuals and communicate insights |

The application should only include workstreams relevant to the supplied assignment.

For example, a descriptive analytics project should not automatically receive a machine learning workstream.

# 6.4 Three Recommended Work Allocations

## Option A: Strength-Based Allocation — Default

The system matches responsibilities to team members' reported strengths.

**Example:**

| Member | Strength | Assigned Workstream |
|---|---|---|
| A | SQL / Python | Data Preparation |
| B | Statistics | Statistical Analysis |
| C | Business Strategy | Research and Recommendations |
| D | Communication | Visualization and Presentation |

**Logic:**

- Prioritize skills relevant to each responsibility.
- Consider stated preferences.
- Avoid substantial workload imbalances where possible.
- Flag responsibilities with no clear skill match.

**Research Justification:** Strength-based allocation received the highest preference in the survey (43.9%).

## Option B: Balanced Workload Allocation

The system distributes estimated work across participants.

**Logic:**

- Estimate approximate effort for each workstream.
- Attempt to minimize large differences in total assigned effort.
- Allow shared ownership of larger tasks.

- Permit manual adjustment.

**Research Justification:** 53.7% reported workload balance as a challenge.

### Option C: Project-Stage Allocation

The system divides responsibilities according to the project's analytical workflow.

Example stages:

1. Problem definition and research
2. Data preparation and analysis
3. Modeling and interpretation
4. Recommendations and presentation

**Logic:**

- Identify major project phases.
- Highlight dependencies between phases.
- Assign responsibility for deliverables within each stage.
- Avoid assuming a single fixed sequence when tasks can run in parallel.

## 6.5 Workload Visibility

Each generated allocation should display estimated relative effort.

**Example:**

| Member | Estimated Effort |
|---|---|
| A | 8 hours |
| B | 10 hours |
| C | 9 hours |
| D | 7 hours |

Estimates are provisional planning assumptions generated from task complexity and scope.

Users can edit estimates, and the interface must clearly distinguish them from measured work hours.

For the initial prototype, a simple effort comparison is sufficient. No sophisticated dashboard is required.

## 6.6 Editable Work Plans

After reviewing the three allocations, the organizer can:

- Reassign responsibilities.
- Rename and edit workstreams.
- Add or remove tasks.
- Adjust estimated effort.
- Select and save the preferred allocation.
- Share the resulting plan with team members.

## 6.7 Functional Requirements

| ID | Requirement | Priority |
|---|---|---|
| FR1.1 | Accept pasted assignment text | P0 |
| FR1.2 | Accept PDF assignment uploads | P0 |
| FR1.3 | Extract deliverables and analytical requirements | P0 |
| FR1.4 | Generate three distinct allocations | P0 |
| FR1.5 | Collect optional member strengths | P0 |
| FR1.6 | Default to strength-based allocation when skill data exists | P0 |
| FR1.7 | Allow editing and saving allocations | P0 |
| FR1.8 | Display approximate workload by member | P1 |
| FR1.9 | Display major task dependencies | P1 |
| FR1.10 | Support additional document formats | P1 |

## 6.8 Acceptance Criteria

The feature is operational when:

1. A user can provide an analytics case prompt.
2. The system correctly identifies the main explicit deliverables.
3. Three distinct allocation options are generated.
4. The skill-based option reflects provided member skills.
5. Users can modify and save a selected allocation.
6. Missing or ambiguous requirements are flagged rather than invented.

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# 7. Core Feature 2: Collaborative Meeting Scheduler

**Priority: P0 — Core MVP**

## 7.1 Feature Description

After selecting a work plan, the organizer generates a shareable meeting availability link.

Team members indicate their available times, and the system identifies overlapping windows suitable for the first team meeting.

The feature should be straightforward, requiring minimal interaction.

## 7.2 User Inputs

- Participant names
- Proposed date range
- Desired meeting duration
- Participant availability

The scheduler should support common meeting durations such as 30, 45, and 60 minutes.

## 7.3 User Workflow

### Step 1: Create Meeting Poll

The organizer selects the proposed meeting dates and duration.

### Step 2: Generate Link

The system creates a unique scheduling link.

### Step 3: Collect Availability

Participants open the link and mark available time blocks.

No account is required for participants.

### Step 4: Identify Overlap

The application calculates which meeting windows accommodate the team.

Results are organized into:

- Full-team availability
- Partial-team availability
- Unavailable or conflicting windows

### Step 5: Confirm Meeting

The organizer selects the preferred meeting window and saves the decision.

The chosen time is displayed on the meeting page.

## 7.4 Analytics-Specific Context

The application may suggest a meeting purpose using the identified analytical project stages.

Examples:

- Initial problem-framing meeting
- Methodology and data discussion
- Analysis review
- Recommendation development

For MVP, these are optional labels rather than automated recurring meetings.

## 7.5 Functional Requirements

| ID | Requirement | Priority |
|---|---|---|
| FR2.1 | Create scheduling poll | P0 |
| FR2.2 | Generate unique shareable link | P0 |
| FR2.3 | Collect availability without participant accounts | P0 |
| FR2.4 | Identify overlapping available times | P0 |
| FR2.5 | Account for selected meeting duration | P0 |
| FR2.6 | Display best partial-overlap options | P0 |
| FR2.7 | Allow participants to revise availability | P0 |

| | | |
|---|---|---|
| FR2.8 | Allow organizer to confirm meeting | P0 |
| FR2.9 | Display and normalize time zones | P0 |
| FR2.10 | Suggest meeting purpose from project stages | P1 |

## 7.6 Acceptance Criteria

The feature is operational when:

1. A user can create a scheduling poll.
2. Multiple participants can submit availability independently.
3. The platform identifies valid overlapping windows.
4. Results reflect the selected meeting duration.
5. The organizer can choose a final meeting time.
6. Time zone differences do not create incorrect overlaps.

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# 8. End-to-End User Journey

### Screen 1: Project Setup

User enters:

- Project name
- Assignment instructions
- Team members
- Optional strengths and preferences

**Action:** Generate Plan

### Screen 2: Assignment Breakdown

The application displays:

- Extracted requirements
- Main deliverables
- Suggested analytical workstreams
- Potential uncertainties in the prompt

**Action:** Review and Continue

### Screen 3: Work Allocation

Three selectable options:

- Strength-Based
- Balanced Workload
- Project-Stage Based

The user can compare and edit the recommended distributions.

**Action:** Save Team Plan

### Screen 4: Meeting Scheduler

Organizer chooses meeting dates and duration.

**Action:** Create Scheduling Link

### Screen 5: Team Availability

Participants enter their availability.

The application identifies overlapping meeting windows.

**Action:** Confirm Meeting

### Final Outcome

The team leaves the platform with:

1. An agreed-upon initial division of responsibilities.
2. A shared understanding of the project's key analytical requirements.
3. A confirmed kickoff meeting time.

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# 9. MVP User Stories

**US1 — Project Interpretation**

As a student, I want to upload my assignment instructions so I can understand the required analytical workstreams without manually organizing every deliverable.

**US2 — Skill-Based Assignment**

As a team organizer, I want to match responsibilities to members' strengths so that the team can use its skills effectively.

**US3 — Workload Fairness**

As a team member, I want to see the approximate effort assigned to each participant so that I can identify potentially uneven workloads.

**US4 — Alternative Work Splits**

As a team organizer, I want to compare different approaches to dividing work so that I can choose the most appropriate structure for the project.

**US5 — Editable Recommendations**

As a team member, I want to modify the AI-generated assignments so that my team retains control over the final plan.

**US6 — Meeting Coordination**

As a student, I want to mark my availability through a link so that I can coordinate without lengthy group chat discussions.

**US7 — Overlap Identification**

As a team organizer, I want to see when everyone is available so that I can select a meeting time quickly.

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# 10. Technical Approach

## AI-Powered Project Interpretation

Use an LLM to interpret assignment instructions and generate structured recommendations.

The processing pipeline should:

1. Extract text from the uploaded instructions.
2. Identify explicit requirements and deliverables.
3. Categorize relevant analytical workstreams.
4. Generate three work allocation strategies.
5. Return editable structured output.

Generated workstreams should be traceable to source requirements where feasible.

The model should distinguish explicit assignment requirements from optional planning suggestions.

## Work Allocation Logic

Use a combination of AI-generated task descriptions and allocation rules.

Possible considerations:

- Task complexity
- Estimated effort
- Team size
- Relevant skills
- Project stages
- Task dependencies

The initial system does not require sophisticated optimization or machine learning models. Simple allocation heuristics are sufficient for validating the concept.

## Meeting Scheduler Logic

Meeting overlap calculations should be deterministic, not generated by an LLM.

The scheduling engine should compare submitted time intervals and determine which windows support the requested meeting duration.

## Data Requirements

Store:

- Project name and instructions
- Extracted deliverables
- Team member names and optional strengths
- Generated and edited work allocations
- Scheduling poll details
- Individual availability
- Confirmed meeting time

## Privacy and Access

Project information should only be available to authorized participants.

Scheduling links should use unpredictable identifiers, participants should not need to create accounts, and the interface should communicate who can see their submitted availability.

Sensitive assignment contents should not be exposed through public project listings.

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# 11. Success Metrics

## North Star Metric: Successful Coordinated Kickoffs

**Definition:** The proportion of started projects in which a team saves an initial work allocation and selects a meeting time.

This metric measures completion of the product's combined value proposition.

## Primary Metrics

| Metric | Definition | Initial Validation Objective |
|---|---|---|
| Work Allocation Completion | Percentage of pilot teams saving a work split | At least 70% |
| Allocation Usefulness | Percentage rating recommendations useful | At least 70% |
| Scheduling Completion | Percentage of polls resulting in a confirmed time | At least 70% |
| Planning Time | Time needed to produce an acceptable work allocation | Improvement over baseline |
| Coordination Time | Time from project kickoff request to agreed work plan and meeting | Target 30% reduction |
| Future Use Intent | Percentage of pilot participants willing to use CaseFlow again | Exceed initial 39% survey interest |

These objectives are hypotheses for validation, not established product performance.

## Secondary Metrics

- Most frequently selected allocation strategy
- Number of manual edits to generated plans
- Perceived fairness of the final allocation
- Number of availability submissions per scheduling poll
- Percentage of meetings with full-team availability
- Reasons participants prefer existing tools

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# 12. MVP Experiment and Validation

## Core Hypothesis

**Teams using CaseFlow AI can produce a useful analytical work allocation and coordinate their first meeting with less effort than teams using their existing planning methods.**

## Experiment 1: Internal Testing

**Participants:** Development team

**Material:** An existing business analytics case competition prompt.

### Procedure

1. Upload the case prompt.
2. Evaluate the extracted requirements.
3. Generate and compare three allocations.
4. Assess the relevance of skill matching.
5. Review and adjust workloads.
6. Test the meeting scheduling process.
7. Record time spent and usability problems.

### Evaluation Questions

- Did the platform identify the required deliverables?
- Were the generated plans meaningfully different?
- Was the skill-based allocation appropriate?
- Were any workload estimates unreasonable?
- Was editing straightforward?
- Did the scheduler correctly identify meeting overlaps?

# Experiment 2: External Student Pilot

**Participants:** Three to five student teams, ideally including teams beyond the development group.

## Procedure

Have teams complete a kickoff exercise using CaseFlow AI and compare the experience with their existing approach.

Where possible, use comparable prompts or counterbalanced exercises to reduce differences in task difficulty.

## Data Collection

**Quantitative:**

- Time to finalize a work allocation
- Number of edits made
- Time or effort required to coordinate availability
- Feature completion rates
- Participant usefulness ratings

**Qualitative:**

- What was most useful?
- Which recommendations were inaccurate?
- Which existing tools would participants continue using?
- What would make them return to the platform?
- Would they recommend the application to another competition team?

## Decision Criteria

**Proceed:** Students successfully complete the workflow, report meaningful benefits, and demonstrate interest in reusing the product.

**Iterate:** Users value work allocation or scheduling but encounter major usability or accuracy problems.

**Reconsider Scope:** Most teams prefer their existing tools and perceive little incremental value from the integrated application.

Because the pilot sample is small, the results will guide the next product iteration rather than serve as a statistically conclusive experiment.

# 13. MVP Development Scope

## P0 — Must Build

| Component | Deliverable |
|---|---|
| Project Input | Paste instructions or upload PDF |
| Team Setup | Enter team member names and skills |
| AI Prompt Interpretation | Identify relevant requirements and deliverables |
| Allocation Generator | Generate three work distributions |
| Allocation Editor | Modify and save work responsibilities |
| Scheduling Poll | Generate shareable link |
| Availability Collection | Allow members to enter available times |
| Overlap Calculation | Display suitable meeting windows |
| Meeting Confirmation | Select and display confirmed meeting |

## P1 — Build If Time Allows

- Relative workload visualization
- Task dependency indicators
- Project-stage meeting suggestions
- Additional document upload formats
- Improved plan-sharing and export options

## P2 — Future Development

- Calendar integrations
- Task progress tracking
- Automated reminders
- Dataset-aware project scoping
- Project analytics dashboards
- Reusable case competition templates

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# 14. Initial Development Timeline

## Week 1 — Functional MVP

**Day 1: Project Setup and Prompt Processing**

Create the project input interface, team member form, and basic assignment interpretation.

**Day 2: Allocation Generation**

Implement the three allocation approaches and structured workstream output.

**Day 3: Editing and Saving**

Develop the work allocation editor and selected-plan storage.

**Day 4: Meeting Scheduler**

Implement scheduling polls, availability submission, and overlap calculation.

**Day 5: Integration and Testing**

Connect the work allocation and scheduling workflows, test using a real case prompt, and resolve critical issues.

The objective is a usable internal prototype, not a production-ready application. If time is limited, P1 enhancements should be deferred.

## Following Week — Validation

Conduct usability sessions with three to five student teams.

Measure completion, time, perceived usefulness, and the value of the combined workflow.

Prioritize changes based on observed behavior rather than requested feature quantity.

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# 15. Risks and Mitigations

| Risk | Mitigation |
|---|---|
| Teams already find dividing work easy | Focus on improving allocation quality rather than merely automating agreement |
| AI produces inaccurate or generic recommendations | Ground workstreams in explicit prompt requirements and allow editing |
| Estimated workloads are unreliable | Use transparent, adjustable estimates |
| Scheduling provides limited improvement | Test convenience against group chats and existing scheduling tools |
| Too many features delay MVP | Restrict development to two core workflows |
| Users prefer ChatGPT plus existing tools | Test whether the integrated experience provides measurable additional value |
| Low adoption intent | Prioritize behavioral validation and repeat-use intent |
| Uploaded assignments contain sensitive information | Limit access and minimize retained data |

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# 16. Competitive Positioning

CaseFlow AI operates at the intersection of three existing tool categories.

| Existing Tool | Primary Function | CaseFlow AI Differentiation |
|---|---|---|
| ChatGPT | General AI assistance | Structured, editable, analytics-aware work allocations integrated with scheduling |
| When2meet / Doodle | Meeting availability | Connects meeting coordination to project planning |
| Notion / Trello / Asana | General project management | Focuses narrowly on the initial project kickoff |

## Positioning Statement

**For analytics students and case competition teams who need to organize collaborative projects, CaseFlow AI is an AI-powered kickoff assistant that combines assignment**

**interpretation, skill-aware work allocation, and meeting scheduling in one lightweight workflow.**

Unlike generic project management tools, it focuses on helping teams get started rather than managing every stage of project execution.

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# 17. Open Questions

The following decisions should be resolved through prototyping and user testing:

1. Should the MVP support anonymous project creation, or require organizer accounts for saving and returning?
2. How much team skill information will students be willing to enter?
3. Should workload be shown in estimated hours or relative effort categories?
4. Should a team be able to use the meeting scheduler without first generating a work allocation?
5. Does combining the two features provide enough value to compete with established tools?
6. Are analytics-specific planning recommendations meaningfully more useful than generic AI-generated task lists?

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# 18. Long-Term Product Vision

The short-term goal is to help analytics teams start projects faster and organize responsibilities more effectively.

Longer-term opportunities include integrating project execution, dataset-aware planning, workload tracking, and analytics-specific collaboration tools.

However, these opportunities should only be pursued after validating the initial kickoff experience.

## Final MVP Definition

**CaseFlow AI transforms an analytics assignment into three editable team work plans and provides a shared scheduling tool to coordinate the first meeting.**

## Product Strategy

**Build for analytics students. Test with case competition teams. Differentiate through skill-aware work allocation and workload fairness. Validate whether the integrated planning and scheduling experience changes user behavior.**